Question2

```r
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✓ dplyr     1.1.4     ✓ readr     2.1.5
## ✓ forcats   1.0.1     ✓ stringr   1.5.2
## ✓ ggplot2   4.0.0     ✓ tibble    3.3.0
## ✓ lubridate 1.9.4     ✓ tidyr     1.3.1
## ✓ purrr     1.1.0
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✕ dplyr::filter() masks stats::filter()
## ✕ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
e errors
```

```r
library(ggthemes)
```

```r
co2 <- read_csv("data/data_raw/owid_co2.csv")
```

```
## New names:
## Rows: 50407 Columns: 80
## ── Column specification
## ──────────────────────────────────────────────────────── Delimiter: "," chr
## (2): country, iso_code dbl (78): ...1, year, population, gdp, cement_co2,
## cement_co2_per_capita, co...
## ℹ Use `spec()` to retrieve the full column specification for this data. ℹ
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## • `` -> `...1`
```

```r
co2_gdp <- co2 |>
  filter(!is.na(iso_code), year >= 1990) |>
  mutate(
    gdp_per_capita = gdp / population
  )
```

```r
baseline_year <- 2010

# Costa Rica 2010 GDP per capita
cr_gdp_2010 <- co2_gdp |>
  filter(country == "Costa Rica", year == baseline_year) |>
  pull(gdp_per_capita) |>
  first()

# Find 5 closest countries by GDP pc in 2010
peer_candidates <- co2_gdp |>
  filter(year == baseline_year,
         !is.na(gdp_per_capita),
         country != "Costa Rica") |>
  mutate(
    gdp_gap = abs(gdp_per_capita - cr_gdp_2010)
  ) |>
  arrange(gdp_gap) |>
  slice_head(n = 5) |>
  pull(country)

peer_countries <- c("Costa Rica", peer_candidates)
peer_countries
```

```
## [1] "Costa Rica"   "Serbia"       "Botswana"     "South Africa" "Barbados"
## [6] "Algeria"
```

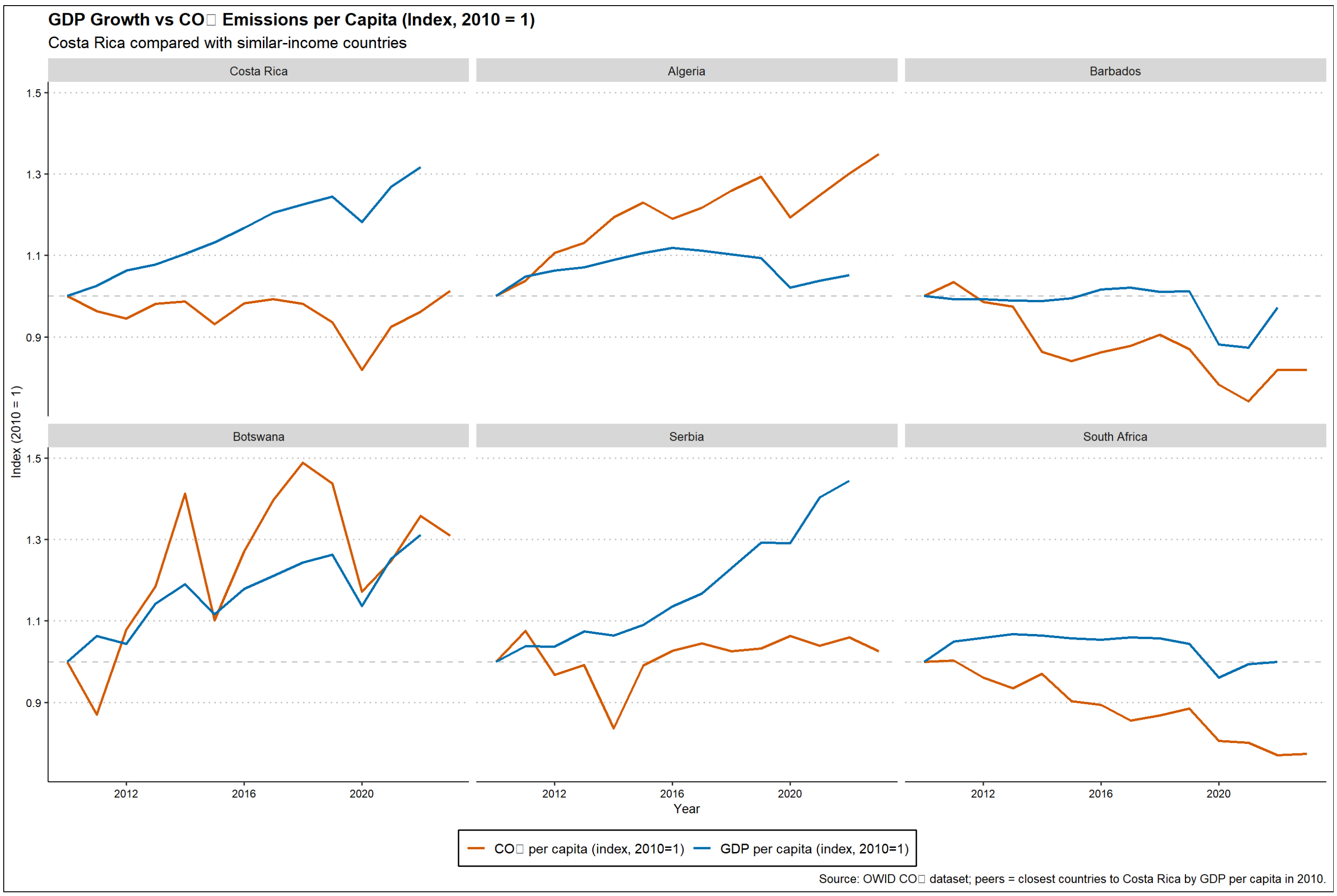
```r
# Baseline values in 2010 for each peer country
base_vals <- co2_gdp |>
  filter(country %in% peer_countries, year == baseline_year) |>
  select(country,
         gdp_pc_base  = gdp_per_capita,
         co2_pc_base  = co2_per_capita)

decouple_df <- co2_gdp |>
  filter(country %in% peer_countries,
         year >= baseline_year, year <= 2023) |>
  left_join(base_vals, by = "country") |>
  mutate(
    gdp_pc_index = gdp_per_capita / gdp_pc_base,
    co2_pc_index = co2_per_capita / co2_pc_base
  )
```

```r
plot_decouple <- decouple_df |>
  select(country, year, gdp_pc_index, co2_pc_index) |>
  pivot_longer(
    cols      = c(gdp_pc_index, co2_pc_index),
    names_to  = "indicator",
    values_to = "index"
  ) |>
  mutate(
    indicator = recode(
      indicator,
      gdp_pc_index = "GDP per capita (index, 2010=1)",
      co2_pc_index = "CO\u2082 per capita (index, 2010=1)"
    )
  )


plot_decouple <- plot_decouple |>
  mutate(
    country = factor(
      country,
      levels = c(
        "Costa Rica", "Algeria", "Barbados", "Botswana", "Serbia", "South Africa")
    )
  )

p2 <- ggplot(plot_decouple,
             aes(x = year, y = index, colour = indicator)) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey80") +
  geom_line(linewidth = 1.0) +
  facet_wrap(~ country, ncol = 3) +
  scale_colour_manual(values = c(
    "GDP per capita (index, 2010=1)" = "#0072B2",
    "CO\u2082 per capita (index, 2010=1)" = "#D55E00"
  )) +
  labs(
    title = "GDP Growth vs CO\u2082 Emissions per Capita (Index, 2010 = 1)",
    subtitle = "Costa Rica compared with similar-income countries",
    x = "Year",
    y = "Index (2010 = 1)",
    colour = NULL,
    caption = "Source: OWID CO₂ dataset; peers = closest countries to Costa Rica by GDP per capi
ta in 2010."
  ) +
  theme_clean(base_size = 12) +
  theme(
    legend.position = "bottom",
    panel.grid.minor = element_blank()
  )
```

```
## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## ℹ Please use the `linewidth` argument instead.
## ℹ The deprecated feature was likely used in the ggthemes package.
##   Please report the issue at <https://github.com/jrnold/ggthemes/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
p2
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```



```
summary_decouple <- decouple_df |>
  group_by(country) |>
  filter(year >= baseline_year) |>
  filter(!is.na(gdp_per_capita), !is.na(co2_per_capita)) |>
  arrange(year, .by_group = TRUE) |>
  summarise(
    start_year     = first(year),
    end_year       = last(year),
    gdp_pc_change  = 100 * (last(gdp_per_capita) / first(gdp_per_capita) - 1),
    co2_pc_change  = 100 * (last(co2_per_capita) / first(co2_per_capita) - 1),
    .groups = "drop"
  )|>
  arrange(desc(gdp_pc_change))

summary_decouple
```

```
## # A tibble: 6 × 5
##   country      start_year end_year gdp_pc_change co2_pc_change
##   <chr>             <dbl>    <dbl>         <dbl>         <dbl>
## 1 Serbia             2010     2022       44.5            5.97
## 2 Costa Rica         2010     2022       31.8           -3.79
## 3 Botswana           2010     2022       31.2           35.8
## 4 Algeria            2010     2022        5.19          30.2
## 5 South Africa       2010     2022       -0.0718       -23.0
## 6 Barbados           2010     2022       -2.82         -18.2
```